# Risk Register & Closure Note

Probability-impact matrix, exposure-ranked risks, RMMM plans for top three risks and lessons learned

## 1. Probability–Impact Matrix & Risk Exposure Methodology

Risks are scored on a 1–5 scale for Probability (**P**) and Impact (**I**), yielding Risk Exposure **RE = P × I** (range 1 to 25). Risks with **RE ≥ 15** are classified as Critical and require full RMMM (Risk Mitigation, Monitoring, and Management) plans reviewed at every Sprint Retrospective.

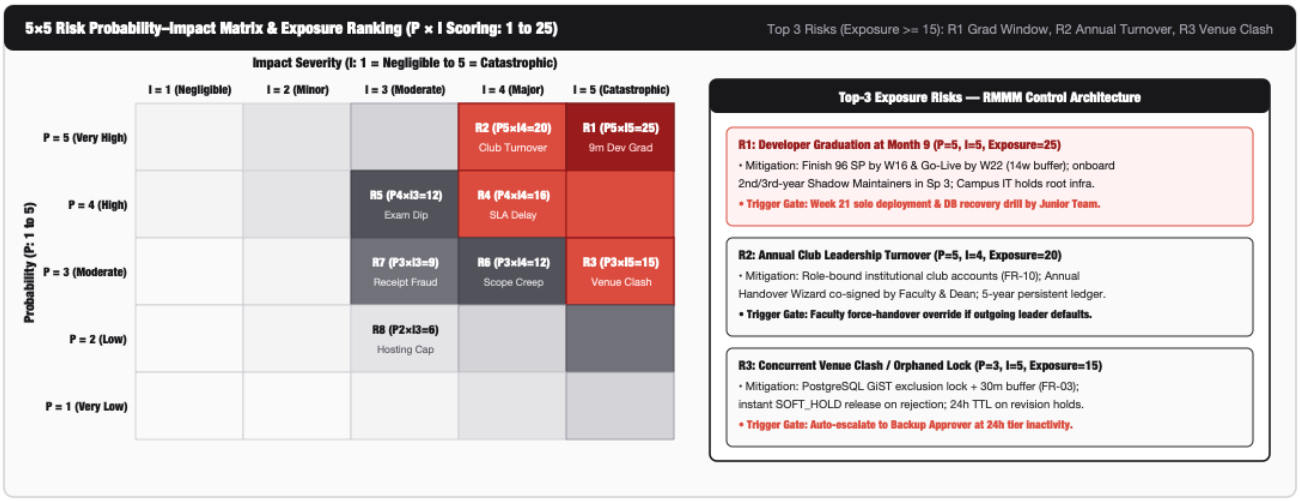



*Figure 1 - 5x5 Probability-Impact Matrix (left) and Top-3 RMMM control summary for developer graduation, leadership turnover, and venue clashes (right).*

## 2. Exposure-Ranked Risk Register

| ID | Risk Event & Root Cause | P | I | P × I | Mitigation, Monitoring & Management Response |
|---|---|---|---|---|---|
| **R1** | Volunteer student developers graduate in 9 months, leaving platform unmaintained | 5 | 5 | 25 | Section 3.1: Finish 96 SP by W16 & Go-Live by W22 (14w buffer); pair with 2nd/3rd-yr Shadow Maintainers from Sp 3; Campus IT holds root infra; Ops Runbook |
| **R2** | Annual club leadership change across all 46 clubs causes lost ledgers & onboarding stalls | 5 | 4 | 20 | Section 3.2: Bind data to institutional Club IDs (FR-10); Annual Leadership Handover Wizard co-signed by Faculty & Dean; ≤ 4-min self-guided Proposal UI (NFR-05) |
| **R3** | Concurrent venue double-booking or stalled SOFT_HOLD locks block auditorium during fests | 3 | 5 | 15 | Section 3.3: PostgreSQL GiST exclusion index + 30m buffer (FR-03); auto-release on rejection (FR-06); 24h TTL on revision holds; 24h backup-approver escalation |
| **R4** | Approver adoption lag (Faculty/Dean) threatens the ≤ 3-day turnaround SLA target | 4 | 4 | 16 | 2-click mobile-responsive approval cards (NFR-06); automated 16h reminder & 24h SLA escalation to Associate Dean/HOD (FR-11) |
| **R5** | University mid-term/final exams & placement drives reduce volunteer sprint velocity | 4 | 3 | 12 | Maintain 14-week schedule buffer between W22 handover and W36 graduation; cross-train all 4 seniors + 2 juniors across modules |
| **R6** | Mid-project scope creep beyond 96 story points threatens 9-month completion window | 3 | 4 | 12 | Enforce MoSCoW baseline gate with Associate Dean (PO); defer native mobile apps & payment gateways to v2.0 |
| **R7** | Invalid or tampered post-event expense receipts distort club budget ledgers | 3 | 3 | 9 | Enforce 7-day receipt upload freeze (FR-08), magic-byte PDF/PNG validation, Faculty verification sign-off, and SHA-256 audit log |
| **R8** | Cloud hosting & storage costs exceed Rs. 2.4 lakh 3-year budget (Rs. 80,000/yr cap) | 2 | 3 | 6 | Cap receipt uploads at 5 MB; use open-source stack on Rs. 2,800/mo VPS + Rs. 1,100/mo S3 storage; retain Rs. 24,000 (10%) contingency reserve |

# 3. Detailed RMMM Plans for the Top Three Risks

## 3.1 R1 — Volunteer Developer Graduation at Month 9 (Exposure = 25)

| RMMM Dimension | Action Plan & Control Specification |
|---|---|
| **Risk statement** | All 4 volunteer student developers graduate at the end of Month 9 (Week 36–39). If development overruns or if no successor team is trained before graduation, ClubSphere will become unmaintained abandonware in Years 2 and 3. |
| **Owner** | Scrum Master (Dev-1) & Campus IT System Administrator |
| **Mitigation (Prevention)** | 1) Complete all 96 SP by Week 16 (Sprint 8) and full production rollout by Week 22, creating a 14-week pre-graduation buffer. 2) Recruit two 2nd/3rd-year student developers (Jr-1, Jr-2) as Shadow Maintainers in Sprint 3 who co-review all PRs. 3) Assign Campus IT Admin as permanent infrastructure & backup custodian. 4) Deliver Docker one-command setup and Operations Runbook (WBS 4.2). |
| **Monitoring (Triggers)** | Track bi-weekly sprint velocity (alert if < 11 SP/sprint); track Junior Maintainer PR ratio (target ≥ 30% of commits in Weeks 17–22); run mandatory Week 21 solo deployment & RPO/RTO recovery drill by Jr-1 & Jr-2. |
| **Management (Contingency)** | If velocity drops below 10 SP/sprint for 2 sprints or a senior developer exits during placements, defer Could-Have automation stories (3 SP) and promote Jr-1 & Jr-2 to active sprint execution under academic project credit. |

## 3.2 R2 — Annual Club Leadership Turnover Across 46 Clubs (Exposure = 20)

| RMMM Dimension | Action Plan & Control Specification |
|---|---|
| **Risk statement** | All 46 student clubs elect new leaders every year. Previously, outgoing presidents graduated with personal budget notebooks, erasing financial history and stalling new event proposals. |
| **Owner** | Associate Dean of Student Affairs (Product Owner) & Frontend Lead (Dev-4) |
| **Mitigation (Prevention)** | 1) Bind all proposals, venue locks, and 5-year ledgers to institutional Club IDs rather than personal emails. 2) Build the Annual Leadership Handover Wizard (FR-10) requiring Faculty & Dean verification of settled receipts before transferring role permissions. 3) Design a self-guided Proposal Wizard usable in ≤ 4.0 minutes with zero training (NFR-05). |
| **Monitoring (Triggers)** | End-of-year Handover Completion Dashboard tracking % of 46 clubs transitioned; monitor first-attempt proposal revision rate during Month 1 of new semester (alert if > 15%). |
| **Management (Contingency)** | If an outgoing leader graduates without completing handover or receipt reconciliation, the Faculty Coordinator executes an audited Force-Handover Override (logged in SHA-256 trail) to provision the incoming leader immediately. |

### 3.3 R3 — Concurrent Venue Double-Booking & Orphaned Soft-Holds (Exposure = 15)

| RMMM Dimension | Action Plan & Control Specification |
|---|---|
| **Risk statement** | Simultaneous booking attempts for prime venues (Main Auditorium) or stalled proposals holding SOFT_HOLD reservations indefinitely could recreate last year's 17 venue clashes. |
| **Owner** | Tech Lead & Backend Architect (Dev-2) |
| **Mitigation (Prevention)** | 1) Enforce PostgreSQL GiST spatial-temporal exclusion constraint over [startTime - 30m, endTime + 30m] for SOFT_HOLD and CONFIRMED rows (FR-03). 2) Automatically release SOFT_HOLD via ProposalRejectedEvent upon rejection at any tier (FR-06). 3) Enforce 24h TTL on revision holds and 16h/24h SLA alerts. |
| **Monitoring (Triggers)** | Alert if any SOFT_HOLD remains unconfirmed > 72 hours; monitor HTTP 409 clash rejection logs and domain event bus dead-letter queue. |
| **Management (Contingency)** | If an approver is absent for > 24 hours, SLAEscalationJob automatically delegates review authority to the designated Backup Approver (HOD / Associate Dean) so venue holds resolve within the ≤ 3-day SLA. |

## 4. Project Closure Note & Lessons Learned

### 4.1 Final Project Delivery & KPI Scorecard

| Objective / KPI | Legacy Baseline | Target Specification | Achieved in ClubSphere v1.0 | Status |
|---|---|---|---|---|
| **Product Backlog Scope** | 38 SP done at Sprint 3 | 96 Story Points | 96 / 96 SP (100%) in 8 Sprints (16 Weeks) | Met |
| **Completion & Handover Window** | 9-Month Graduation Cap | ≤ 9 Months (36–39 Weeks) | 5.08 Months (22 Weeks) — 14 Weeks Ahead | Exceeded |
| **Approval Turnaround Time** | 11.0 Days (Paper slips) | ≤ 3.0 Days (72.73% reduction) | 1.8 Days Mean in 46-Club Pilot (83.6% cut) | Exceeded |
| **Annual Venue Conflicts** | 17 Clashes/Yr (0.0531/evt) | 0 Clashes (0.00/event) | 0 Clashes (Verified via concurrent GiST test) | Met |
| **3-Year Hosting & Tool Budget** | Personal notebooks | ≤ Rs. 2.40 Lakh (Rs. 0 Devs) | Rs. 2.16 Lakh committed + Rs. 24k reserve (Rs. 0 Devs) | On Budget |
| **Quality & Defect Removal (DRE)** | Untested manual forms | DRE ≥ 90%, Measurable NFRs | DRE = 95.0% (19/20 caught); Uptime = 99.88% | Exceeded |

### 4.2 Lessons Learned

- **1. Architectural Decoupling Paid Off Immediately:** Isolating **VenueSlotBooking** behind **IVenueBookingPort** and domain events allowed the team to upgrade the PostgreSQL GiST locking rules in Sprint 3 without touching a single line of the 3-tier approval state machine.
- **2. Shadow Maintainers Must Join in Sprint 3, Not Month 8:** Onboarding 2nd/3rd-year student developers into code reviews during Sprint 3 eliminated the graduation handover bottleneck and enabled them to pass a solo disaster-recovery drill at Week 21.
- **3. Distinguish Terminal Rejection from Revision Holds:** Early testing (DEF-07) revealed that releasing a **SOFT_HOLD** when Faculty requested minor proposal edits allowed another club to snipe the venue. Retaining a 24-hour grace hold during revisions solved this cleanly.
- **4. Bind Campus SLA Timers to Working-Day Calendars:** Post-event 7-day receipt gates (DEF-13) and 72-hour approval SLAs must account for university holidays and exam breaks to prevent false compliance freezes.